Nonlinear Analysis: Hybrid Systems 50 (2023) 101378

Contents lists available at ScienceDirect

Nonlinear Analysis: Hybrid Systems

journal homepage: www.elsevier.com/locate/nahs

On controllability and observability of a class of fractional-order switched systems with impulse ☆

Jiayuan Yan [a,b], Bin Hu [c,*], Zhi-Hong Guan [a], Tao Li [d], Ding-Xue Zhang [e]

[a] School of Artificial Intelligence and Automation, Huazhong University of Science and Technology, Wuhan 430074, PR China
[b] School of Artificial Intelligence, Henan University, Zhengzhou 450000, PR China
[c] School of Future Technology, South China University of Technology, and Pazhou Lab, Guangzhou, 510000, PR China
[d] School of Electrical Engineering and Automation, Hubei Normal University, Huangshi 435000, PR China
[e] School of Petroleum Engineering, Yangtze University, Jingzhou 434000, PR China

ARTICLE INFO

Article history:
Received 2 July 2022
Received in revised form 11 April 2023
Accepted 18 May 2023
Available online 9 June 2023

Keywords:
Controllability
Observability
Fractional-order switched system
Impulse

ABSTRACT

This paper is devoted to investigating two structural properties for a class of fractional-order switched and impulsive systems. The structural properties, i.e., observability and controllability, are explored mainly based on an algebraic approach. More specifically, firstly, according to the Laplace transform and mathematical induction, the general solution of such hybrid fractional-order systems is obtained over every impulsive interval. Next, applying the solution derived and relevant matrix theory, several necessary and sufficient controllability and observability criteria that take the form of a row of Gramian matrices are analytically established in terms of a deterministic impulse-switching time sequence. Resorting to the property of matrix Mittag-Leffler function, the developed controllability and observability Gramian criteria are further converted to some easy-test Kalman-type rank conditions. Finally, a numerical example illustrating the theoretical controllability and observability conditions is given.

1. Introduction

Nowadays, fractional calculus has become a powerful tool in the modeling and control of plenty of practical systems, such as fractional-order circuit [1,2], fractional PID control [3], fractional-order biological network [4] and so forth. This motivates the extensive investigations on several important topics for fractional-order systems, including stability [5], tracking [6], observability and controllability [7,8].

Among these research problems, controllability and observability are two important structural properties, which are strongly related to many aspects within systems and control, e.g., optimal control, pole assignment, observer design and structural decomposition. The former is the ability that, under an admissible control over a finite time interval, the state of the system can be driven from an arbitrary point to another arbitrary point. The latter means an ability of the system that the initial state will be uniquely recovered by the measured output and known or unknown input in a finite time interval. During the past years, controllability and observability have been two important research topics in both integer-order and fractional-order systems. In terms of fractional-order systems, numerous studies on observability and controllability

☆ This work was supported in part by the National Natural Science Foundation of China under Grants 61976100, 62233007 and 61976099. and by the Guangzhou Key Field Artificial Intelligence Social Experiment Research under Fund PZL2023ZZ0001.

* Corresponding author.
E-mail address: huu@scut.edu.cn (B. Hu).

https://doi.org/10.1016/j.nahs.2023.101378

have been systematically developed in [7–17]. Specifically, for fractional-order discrete-time systems, controllability and observability were investigated in terms of linear [7] and nonlinear [8] cases, respectively. The controllability was studied for fractional-order systems with state delay [9,10] and input delay [11,12]. Furthermore, the observability and/or controllability tests for fractional-order systems with different impulsive effects can also be found in [13–17].

On the other hand, switched system, a typical hybrid system, is characterized by several subsystems and a switching law [18]. Over the past years, plenty of theoretical problems have been extensively investigated for switched systems with integer order derivative [19–29]. For example, the stability and stabilization were investigated for a hybrid system under inappropriate impulsive switching signals in [23]. A geometric approach was adopted to establish the controllability and observability conditions for a class of linear impulsive and switching systems in [24] by supposing nonsingularity of impulsive reset matrices, while, in [28], the same structural properties have been analyzed without the nonsingularity assumption. In [25], the observability and controllability properties for linear time-variant impulsive and switching systems were studied by means of an algebraic tool. Different from the above mentioned integer-order switched systems, fractional switched systems can be used to describe some continuous-discrete dynamics with memory and historical-dependent processes. The important motivation for investigating fractional switched systems lies two parts: first, many practical dynamical processes involving fractional dynamics proverbially undergo different submodes, e.g., fractional switched multi-robot networks [30] and fractional switched circuit systems [31]; second, switching among fractional-order systems usually generates some essential effects on the system dynamics including stability and stabilization and so forth [32,33]. Recent years, many meaningful results about such systems have been established. For instance, the fault estimation and accommodation method has been proposed for nonlinear fractional switched systems which can reveal the tradeoff between the switching law, submode dynamics, and value of fractional order [34]. The authors studied the exponential stability problem for a class of switched fractional systems, where switching and impulse signals were introduced asynchronously [35]. By considering the time-delay effect in a fractional switched system, several stability criteria that allow some unstable subsystems have been established [36]. Moreover, it can be found that the impulsive effect, a common phenomena in real-world models, has also been introduced in the fractional switched systems to model some complex dynamics by a number of scholars [37,38].

It should be noted that most of the above mentioned works mainly study the stability and stabilization of fractional switched systems. There has not been relevant investigation on controllability and observability for a fractional switched system in the existing publications, especially considering the impulsive effect. Hence, it is necessary and important to explore some key controllability and observability conditions for a kind of fractional-order switched and impulsive systems (FOSISs). For integer switched and/or impulsive systems, controllability and observability conditions have been established in [24,25,28,29,39,40]. However, the analytical methods and results cannot be applied to cope with the structural properties of FOSISs directly. Compared to these integer cases, the study on controllability and observability of FOSISs needs to overcome more difficulties. The first is how to establish an explicit expression of the solution for a FOSIS since one has to cope with the memory property of fractional derivative and impulsive and switching effects. Besides, the state transition matrix in a fractional system cannot satisfy some classical properties compared to that in an integer system. Thus, it is important to establish some easily-verified controllability and observability criteria avoiding using those properties. The last is how to develop some useful results for Mittag-Leffler matrix function for transferring the controllability and observability conditions in the form of a row of Gramian matrices to a Kalman-type one.

Motivated by the above observations, in the current work, the controllability and observability conditions of FOSISs are established for the first time. It is worthy to point out the following contributions. First, a new FOSIS model is introduced by updating the lower limit in fractional derivative and the formula for an explicit solution of this class of FOSISs is presented with respect to an admissible switching function by Laplace transform and mathematical induction. Then, by using the property of positive semi-definite matrix, several new necessary and sufficient algebraic controllability and observability criteria that take the form of a row of Gramian matrices are analytically established based on the obtained system solution. These controllability and observability criteria are further converted to some easy-test Kalman-type rank conditions by establishing the properties of Mittag-Leffler matrix function. The established results imply that the switching rule and impulsive control may affect the structural properties of FOSISs. Finally, an example with three different subsystems is given to illustrate the developed theory.

The rest of this paper is arranged as follows. In Section 2, a model of FOSISs is introduced and some important preliminary lemmas are shown. In Section 3, the explicit solution of FOSISs is obtained and based on this, several algebraic controllability and observability tests are further established. A numerical example is provided to confirm the validity of the theoretical founding in Section 4. Section 5 gives some concluding remarks.

Notations : Let $\mathbb{N}$ and $\mathbb{R}$ be respectively the sets of positive integers and real numbers. The superscript T represents the transpose of a vector or matrix. Denote $\mathbb{N}_i^k = \{i, i+1, i+2, \ldots, k\}$, where $i, k \in \mathbb{N}$ and $i \leq k$. I represents the identity matrix with appropriate dimension.

2. Modeling and lemmas

In this paper, we consider a FOSIS described by the following model:

$$ {}_{t_{k-1}}^{C}D_t^{\alpha} x(t) = A_{r(t)}x(t) + B_{r(t)}u(t), \quad t \in [t_0, t_f]\backslash\{t_k\}_{k=1}^{M-1}, $$

$$\begin{aligned} \Delta x(t) &= E_{r(t)}x(t) + F_{r(t)}u_k, \quad t = t_k, \\ x(t_0) &= x_0, \\ y(t) &= C_{r(t)}x(t) + D_{r(t)}u(t), \end{aligned} \tag{2.1}$$

where $x(t) \in \mathbb{R}^n$, $u(t) \in \mathbb{R}^d$, and $y(t) \in \mathbb{R}^p$ respectively represent the state, input, and output vectors. $u_k = u(t_k)$ is a discrete control. For $t \in (t_{k-1}, t_k]$, the switching function satisfies $r(t) \equiv r(t_{k-1}^+) = r(t_k) \in \mathbb{N}_1^q$, where t_{k-1} represents the switching instant and q is the number of the subsystems. $A_i, B_i, C_i, D_i, E_i, F_i$ are the time-invariant system matrices with appropriate dimensions, where $i \in \mathbb{N}_1^q$. $\Delta x(t_k) = x(t_k^+) - x(t_k^-)$ is the impulsive jump of state, where $\lim_{s\to 0^+} x(t_k+s) = x(t_k^+)$ and $\lim_{s\to 0^+} x(t_k - s) = x(t_k^-)$. To avoid the occurrence of accumulation points, assume that the switching (or impulsive) instants satisfy $0 = t_0 < t_1 < t_2 < \cdots < t_k < \cdots < t_{M-1} < t_M = t_f$, and $t_k - t_{k-1} > t_{min}$, where $M \in \mathbb{N}$ and t_{min} is a fixed positive constant. Besides, every solution of (2.1) is assumed to be left-continuous at t_k, i.e., $x(t_k^-) = x(t_k)$. x_0 is the initial state of (2.1). Let h_k denote the length of the kth dwell time, i.e., $h_k = t_k - t_{k-1}$, where $k \in \mathbb{N}$. ${}_{t_{k-1}}^{C}D_t^{\alpha}$ is the Caputo fractional derivative [2] with order α $(0 < \alpha < 1)$, where the lower limit is t_{k-1}. The representation of ${}_{t_{k-1}}^{C}D_t^{\alpha}$ is given by

$${}_{t_{k-1}}^{C}D_t^{\alpha}x(t) = \frac{1}{\Gamma(1-\alpha)}\int_{t_{k-1}}^{t}(t-s)^{-\alpha}\dot{x}(s)\mathrm{d}s,$$

where $\Gamma(\cdot)$ is Euler's Gamma Function satisfying

$$\Gamma(z) = \int_0^{\infty} e^{-t}t^{z-1}\mathrm{d}t, \ \ \mathrm{Re}(z) > 0.$$

Remark 2.1. In some existing literature [30,31,33,34], the fractional switched systems with fixed lower limit of fractional derivative is investigated (the lower limit usually is assumed to be zero). However, in system (2.1), the lower limit needs to be updated at every switched (impulsive) instant. It is not denied that fractional calculus is affected by the previous process at any moment. There are two main reasons considering the updated lower limit, that is the theoretical significance and engineering realization. (1) When investigating the controllability and observability of a linear hybrid system, the explicit system solution is very important. If the lower limit is not updated, the history from the initial time will be considered when the system evolves over some dwell time. Then one will have to cope with the integral related to the solution of system in all past impulsive intervals. This will result in much difficulty in establishing the explicit solution representation of fractional linear switched systems. To deal with this problem, we update t_{k-1} at switching instants, and thus the integrals of the solution in past impulsive intervals are avoided. Hence, the solution to the fractional switched systems can be constructed by linking the solution on every interval (see Lemma 3.1). (2) On other hand, from the view of engineering realization, in the fractional switched systems (2.1), the switching rule $r(t)$ is a regulation which regulates the updates of lower limit t_{k-1} of fractional order derivative and right matrices $A_{r(t)}, B_{r(t)}, C_{r(t)}, D_{r(t)}, E_{r(t)}, F_{r(t)}$. It is reasonable to define a switching-law-regulated fractional switched system and do some related researches about this kind of systems. Thus, it is not difficult to realize this system in the engineering. In fact, in the study of fractional switched and/or impulsive systems, some researchers have also investigated the updated lower limit of fractional derivative, e.g., fractional switched systems [32], fractional impulsive systems [41,42], fractional switched systems with impulse [36]. For more explanations about the updated lower limit, one can refer to these related papers.

Definition 2.1 (*[2]*)**.** The definition of Mittag-Leffler function with two parameters is given by

$$\mathcal{E}_{\alpha,\beta}(z) = \sum_{i=0}^{\infty}\frac{z^i}{\Gamma(i\alpha+\beta)}, \quad z \in \mathbb{C}, \quad \alpha > 0, \quad \beta > 0,$$

which yields

$$\frac{\mathrm{d}^n}{\mathrm{d}z^n}\mathcal{E}_{\alpha,\beta}(z) = \sum_{i=0}^{\infty}\frac{(i+n)!z^i}{i!\Gamma(i\alpha+n\alpha+\beta)}.$$

In particular, when $\beta = 1$, the above Mittag-Leffler function is reduced to $\mathcal{E}_{\alpha,1}(z) := \mathcal{E}_{\alpha}(z) = \sum_{i=0}^{\infty}\frac{z^i}{\Gamma(i\alpha+1)}$, which is called one parameter Mittag-Leffler function. It can be found that $\mathcal{E}_{\alpha}(z)$ is in fact an extension of $\mathcal{E}_1(z) = e^z$.

In [1], the result similar to Lemma 2.1 has been proved for one parameter Mittag-Leffler function (i.e., $\mathcal{E}_{\alpha,1}(Xt^{\alpha})$) when X has n distinct eigenvalues. The Lemma 2.1 will give the proof for $\mathcal{E}_{\alpha,\alpha}(Xt^{\alpha})$ when X has same eigenvalues.

Lemma 2.1. *Assume that $X \in \mathbb{R}^{n\times n}$. Then there exist n functions $f_i(t)$, $i = 0, 1, 2, \ldots, n-1$, which are continuous and linearly independent such that*

$$\mathcal{E}_{\alpha,\alpha}(Xt^{\alpha}) = \sum_{i=0}^{n-1}f_i(t)X^i, \quad t \geq 0.$$

Proof. According to Cayley–Hamilton Theorem, $\mathcal{E}_{\alpha,\alpha}(Xt^{\alpha})$ has the following expression:

$$\mathcal{E}_{\alpha,\alpha}(Xt^{\alpha}) = \sum_{i=0}^{\infty} \frac{X^i t^{i\alpha}}{\Gamma(i\alpha+\alpha)} = \sum_{i=0}^{n-1} f_i(t)X^i,$$

where $f_i(t)$ is a continuous function to be obtained. It is known that the eigenvalues of X also satisfy the above equation. When the matrix X has n distinct eigenvalues, this lemma can be proved similar to [1]. Next, it will be shown that this lemma still holds when X has same eigenvalues.

Without loss of generality, assume that λ_1 has multiplicity 2 and $\lambda_1, \lambda_3, \ldots, \lambda_n$ are distinct eigenvalues. Then we have

$$\mathcal{E}_{\alpha,\alpha}(\lambda_1 t^{\alpha}) = \sum_{i=0}^{n-1} f_i(t)\lambda_1^i, \mathcal{E}_{\alpha,\alpha}(\lambda_3 t^{\alpha}) = \sum_{i=0}^{n-1} f_i(t)\lambda_3^i, \ldots, \mathcal{E}_{\alpha,\alpha}(\lambda_n t^{\alpha}) = \sum_{i=0}^{n-1} f_i(t)\lambda_n^i.$$

Taking the derivative of the first equation with respect to λ_1, we obtain a compact form as follows

$$\begin{bmatrix} \mathcal{E}_{\alpha,\alpha}(\lambda_1 t^{\alpha}) \\ \frac{\partial \mathcal{E}_{\alpha,\alpha}(\lambda_1 t^{\alpha})}{\partial \lambda_1} \\ \mathcal{E}_{\alpha,\alpha}(\lambda_3 t^{\alpha}) \\ \vdots \\ \mathcal{E}_{\alpha,\alpha}(\lambda_n t^{\alpha}) \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & \lambda_1 & \lambda_1^2 & \cdots & \lambda_1^{n-1} \\ 0 & 1 & 2\lambda_1 & \cdots & (n-1)\lambda_1^{n-2} \\ 1 & \lambda_3 & \lambda_3^2 & \cdots & \lambda_3^{n-1} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & \lambda_n & \lambda_n^2 & \cdots & \lambda_n^{n-1} \end{bmatrix}}_{\bar{V}} \begin{bmatrix} f_0(t) \\ f_1(t) \\ f_2(t) \\ \vdots \\ f_{n-1}(t) \end{bmatrix}.$$

Thus, we have

$$\begin{bmatrix} f_0(t) \\ f_1(t) \\ f_2(t) \\ \vdots \\ f_{n-1}(t) \end{bmatrix} = \bar{V}^{-1} \begin{bmatrix} \mathcal{E}_{\alpha,\alpha}(\lambda_1 t^{\alpha}) \\ \frac{\partial \mathcal{E}_{\alpha,\alpha}(\lambda_1 t^{\alpha})}{\partial \lambda_1} \\ \mathcal{E}_{\alpha,\alpha}(\lambda_3 t^{\alpha}) \\ \vdots \\ \mathcal{E}_{\alpha,\alpha}(\lambda_n t^{\alpha}) \end{bmatrix}.$$

Hence, the functions $f_i(t)$ can be obtained. Next, we prove that $f_i(t)(i = 0, 1, 2, \ldots, n-1)$ are linearly independent. We assume that these functions $f_i(t)(i = 0, 1, 2, \ldots, n-1)$ are linearly dependent. Then, the following equation

$$z \begin{bmatrix} f_0(t) \\ f_1(t) \\ \vdots \\ f_{n-1}(t) \end{bmatrix} = 0,$$

where $z = [z_1, z_2, \ldots, z_n]$ is a not trivial complex vector. Then

$$z\bar{V}^{-1} \begin{bmatrix} \mathcal{E}_{\alpha,\alpha}(\lambda_1 t^{\alpha}) \\ \frac{\partial \mathcal{E}_{\alpha,\alpha}(\lambda_1 t^{\alpha})}{\partial \lambda_1} \\ \mathcal{E}_{\alpha,\alpha}(\lambda_3 t^{\alpha}) \\ \vdots \\ \mathcal{E}_{\alpha,\alpha}(\lambda_n t^{\alpha}) \end{bmatrix} = 0.$$

It is obtained that $\{\mathcal{E}_{\alpha,\alpha}(\lambda_1 t^{\alpha}), \frac{\partial \mathcal{E}_{\alpha,\alpha}(\lambda_1 t^{\alpha})}{\partial \lambda_1}, \mathcal{E}_{\alpha,\alpha}(\lambda_3 t^{\alpha}), \ldots, \mathcal{E}_{\alpha,\alpha}(\lambda_n t^{\alpha})\}$ are linear dependent. Then $p_1\mathcal{E}_{\alpha,\alpha}(\lambda_1 t^{\alpha}) + p_2\frac{\partial \mathcal{E}_{\alpha,\alpha}(\lambda_1 t^{\alpha})}{\partial \lambda_1} + p_3\mathcal{E}_{\alpha,\alpha}(\lambda_3 t^{\alpha}) + \cdots + p_n\mathcal{E}_{\alpha,\alpha}(\lambda_n t^{\alpha}) = 0$ holds for a series of not trivial constants $p_1, p_2, \ldots, p_n$. According to the definition of the Mittag-Leffler function, the above equation becomes $\sum_{k=0}^{\infty} \frac{t^{k\alpha}}{\Gamma(k\alpha+\alpha)}(p_1\lambda_1^k + p_2k\lambda_1^{k-1} + p_3\lambda_3^k + \cdots + p_n\lambda_n^k) = 0$. According to Lemma 3.3 in [1], the fractional order power function $\{1, t^{\alpha}, t^{2\alpha}, \ldots, t^{(k-1)\alpha}\}$ are linearly independent. Then $p_1\lambda_1^k + p_2k\lambda_1^{k-1} + p_3\lambda_3^k + \cdots + p_n\lambda_n^k = 0, \forall k$. Hence, $\bar{V}^T[p_1, p_2, \ldots, p_n]^T = 0$. Since $\bar{V}$ has full rank, we conclude that $p_1 = p_2 = \cdots = p_n = 0$, which contradicts the assumption that $p_1, p_2, \ldots, p_n$ are not trivial. Thus Lemma 2.1 also holds if matrix X has same eigenvalues. ■

Lemma 2.2 ([43]). *Let $A_1, A_2, \ldots, A_M$ be $n \times n$ positive semi-definite matrices. Assume that $P = (A_1, A_2, \ldots, A_M)$ and $Q = \sum_{i=1}^{M} A_i$, then rank(P) =rank(Q).*

3. Controllability and observability for FOSISs

In this section, we firstly obtain the general solution representation of (2.1) by iteration, and then systematically analyze the controllability and observability of (2.1).

3.1. General solution of FOSISs

Before investigating the controllability and observability of system (2.1), the solution of system (2.1) on the finite interval $[t_0, t_M]$ ($M \in \mathbb{N}$) will be established. For convenience, the time sequence occurring switching and impulse on $[t_0, t_M]$ is given by $\Sigma := \{t_i\}_{i=1}^M$.

Lemma 3.1. *Given an impulse-switching time sequence $\Sigma := \{t_i\}_{i=1}^M$, the general solution representation of (2.1) is explicitly given by the following formula*

$$x(t) = \begin{cases} \mathcal{E}_\alpha(A_{r(t_1)}(t-t_0)^\alpha)x_0 + \displaystyle\int_{t_0}^{t}(t-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_1)}(t-s)^\alpha)B_{r(t_1)}u(s)\mathrm{d}s, \quad \forall t \in [t_0, t_1], \\ \mathcal{E}_\alpha(A_{r(t_k)}(t-t_{k-1})^\alpha)\Big\{ \mathit{\Pi}_{j=k-1}^{1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)x_0 \\ +\displaystyle\sum_{i=1}^{k-1}\mathit{\Pi}_{j=k-1}^{i+1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)(I+E_{r(t_i)}) \\ \times\displaystyle\int_{t_{i-1}}^{t_i}(t_i-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}(t_i-s)^\alpha)B_{r(t_i)}u(s)\mathrm{d}s \\ +\displaystyle\sum_{i=2}^{k-1}\mathit{\Pi}_{j=k-1}^{i}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)F_{r(t_{i-1})}u_{i-1} + F_{r(t_{k-1})}u_{k-1}\Big\} \\ +\displaystyle\int_{t_{k-1}}^{t}(t-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_k)}(t-s)^\alpha)B_{r(t_k)}u(s)\mathrm{d}s, \quad \forall t \in (t_{k-1}, t_k], \quad k = 2, 3, \ldots, M. \end{cases}$$

Proof. Similar to [1], applying the Laplace transform for system (2.1), the system solution on the time interval $[t_0, t_1]$ is given by

$$x(t) = \mathcal{E}_\alpha(A_{r(t_1)}(t-t_0)^\alpha)x_0 + \int_{t_0}^{t}(t-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_1)}(t-s)^\alpha)B_{r(t_1)}u(s)\mathrm{d}s,$$

which implies that

$$x(t_1) = \mathcal{E}_\alpha(A_{r(t_1)}h_1^\alpha)x_0 + \int_{t_0}^{t_1}(t_1-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_1)}(t_1-s)^\alpha)B_{r(t_1)}u(s)\mathrm{d}s.$$

According to the representation of state jump on t_1, one can obtain

$$\begin{aligned} x(t_1^+) = (I+E_{r(t_1)})\Big[\, &\mathcal{E}_\alpha(A_{r(t_1)}h_1^\alpha)x_0 \\ &+ \int_{t_0}^{t_1}(t_1-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_1)}(t_1-s)^\alpha)B_{r(t_1)}u(s)\mathrm{d}s\Big] + F_{r(t_1)}u_1. \end{aligned}$$

Similarly, when $t \in (t_1, t_2]$, it yields

$$x(t) = \mathcal{E}_\alpha(A_{r(t_2)}(t-t_1)^\alpha)x(t_1^+) + \int_{t_1}^{t}(t-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_2)}(t-s)^\alpha)B_{r(t_2)}u(s)\mathrm{d}s.$$

Substituting $x(t_1^+)$ into the above equation, one obtains

$$\begin{aligned} x(t) = \mathcal{E}_\alpha(A_{r(t_2)}(t-t_1)^\alpha)\Big\{ &(I+E_{r(t_1)})\Big[\,\mathcal{E}_\alpha(A_{r(t_1)}h_1^\alpha)x_0 \\ &+ \int_{t_0}^{t_1}(t_1-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_1)}(t_1-s)^\alpha)B_{r(t_1)}u(s)\mathrm{d}s\Big] + F_{r(t_1)}u_1\Big\} \\ &+ \int_{t_1}^{t}(t-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_2)}(t-s)^\alpha)B_{r(t_2)}u(s)\mathrm{d}s, \end{aligned}$$

which implies that $k = 2$ holds. Assume that this lemma is satisfied for $t \in (t_{k-1}, t_k]$. Thus, we can derive the following equation

$$x(t) = \mathcal{E}_\alpha(A_{r(t_k)}(t-t_{k-1})^\alpha)\Big\{ \mathit{\Pi}_{j=k-1}^{1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)x_0$$

$$
\begin{aligned}
&+\sum_{i=1}^{k-1}\Pi_{j=k-1}^{i+1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)(I+E_{r(t_i)})\int_{t_{i-1}}^{t_i}(t_i-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}(t_i-s)^\alpha)B_{r(t_i)}u(s)\mathrm{d}s\\
&+\sum_{i=2}^{k-1}\Pi_{j=k-1}^{i}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)F_{r(t_{i-1})}u_{i-1}+F_{r(t_{k-1})}u_{k-1}\Bigg\}\\
&+\int_{t_{k-1}}^{t}(t-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_k)}(t-s)^\alpha)B_{r(t_k)}u(s)\mathrm{d}s. \qquad (3.2)
\end{aligned}
$$

Note that when $t\in(t_k,t_{k+1}]$, $x(t)$ has the following expression:

$$
x(t)=\mathcal{E}_\alpha(A_{r(t_{k+1})}(t-t_k)^\alpha)x(t_k^+)+\int_{t_k}^{t}(t-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_{k+1})}(t-s)^\alpha)B_{r(t_{k+1})}u(s)\mathrm{d}s,
$$

where $x(t_k^+)$ is calculated by (3.2) as follows

$$
\begin{aligned}
x(t_k^+)=\;&(I+E_{r(t_k)})\mathcal{E}_\alpha(A_{r(t_k)}h_k^\alpha)\Bigg\{\Pi_{j=k-1}^{1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)x_0\\
&+\sum_{i=1}^{k-1}\Pi_{j=k-1}^{i+1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)(I+E_{r(t_i)})\int_{t_{i-1}}^{t_i}(t_i-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}(t_i-s)^\alpha)B_{r(t_i)}u(s)\mathrm{d}s\\
&+\sum_{i=2}^{k-1}\Pi_{j=k-1}^{i}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)F_{r(t_{i-1})}u_{i-1}+F_{r(t_{k-1})}u_{k-1}\Bigg\}\\
&+(I+E_{r(t_k)})\int_{t_{k-1}}^{t_k}(t_k-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_k)}(t_k-s)^\alpha)B_{r(t_k)}u(s)\mathrm{d}s+F_{r(t_k)}u_k.
\end{aligned}
$$

Then

$$
\begin{aligned}
x(t)\;&=\mathcal{E}_\alpha(A_{r(t_{k+1})}(t-t_k)^\alpha)(I+E_{r(t_k)})\mathcal{E}_\alpha(A_{r(t_k)}h_k^\alpha)\Bigg\{\Pi_{j=k-1}^{1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)x_0\\
&+\sum_{i=1}^{k-1}\Pi_{j=k-1}^{i+1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)(I+E_{r(t_i)})\int_{t_{i-1}}^{t_i}(t_i-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}(t_i-s)^\alpha)B_{r(t_i)}u(s)\mathrm{d}s\\
&+\sum_{i=2}^{k-1}\Pi_{j=k-1}^{i}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)F_{r(t_{i-1})}u_{i-1}+F_{r(t_{k-1})}u_{k-1}\Bigg\}\\
&+\mathcal{E}_\alpha(A_{r(t_{k+1})}(t-t_k)^\alpha)(I+E_{r(t_k)})\int_{t_{k-1}}^{t_k}(t_k-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_k)}(t_k-s)^\alpha)B_{r(t_k)}u(s)\mathrm{d}s\\
&+\mathcal{E}_\alpha(A_{r(t_{k+1})}(t-t_k)^\alpha)F_{r(t_k)}u_k+\int_{t_k}^{t}(t-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_{k+1})}(t-s)^\alpha)B_{r(t_{k+1})}u(s)\mathrm{d}s\\
&=\mathcal{E}_\alpha(A_{r(t_{k+1})}(t-t_k)^\alpha)\Bigg\{\Pi_{j=k}^{1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)x_0+\sum_{i=1}^{k}\Pi_{j=k}^{i+1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)(I+E_{r(t_i)})\\
&\times\int_{t_{i-1}}^{t_i}(t_i-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}(t_i-s)^\alpha)B_{r(t_i)}u(s)\mathrm{d}s+\sum_{i=2}^{k}\Pi_{j=k}^{i}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)F_{r(t_{i-1})}u_{i-1}\\
&+F_{r(t_k)}u_k\Bigg\}+\int_{t_k}^{t}(t-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_{k+1})}(t-s)^\alpha)B_{r(t_{k+1})}u(s)\mathrm{d}s.
\end{aligned}
$$

Hence, the general solution holds for $t\in(t_k,t_{k+1}]$ $(k\in\mathbb{N})$. ■

Remark 3.1. By updating the lower limit of the fractional derivative and setting the switching instants as impulsive instants, one can obtain the solution of FOSISs by Laplace transform and mathematical induction. Although the solution on every subinterval that is dependent on Mittag-Leffler matrix function is established similar to the integer order systems, the subsequent controllability and observability conditions are still different from the existing integer order case.

According to Lemma 3.1, when $t=t_M$, one has the following

$$
x(t_M)=\mathcal{E}_\alpha(A_{r(t_M)}h_M^\alpha)\Bigg\{\Pi_{j=M-1}^{1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)x_0
$$

$$
\begin{aligned}
&+\sum_{i=1}^{M-1}\Pi_{j=M-1}^{i+1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)(I+E_{r(t_i)})\\
&\times\int_{t_{i-1}}^{t_i}(t_i-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}(t_i-s)^\alpha)B_{r(t_i)}u(s)\mathrm{d}s\\
&\left.+\sum_{i=2}^{M-1}\Pi_{j=M-1}^{i}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)F_{r(t_{i-1})}u_{i-1}+F_{r(t_{M-1})}u_{M-1}\right\}\\
&+\int_{t_{M-1}}^{t_M}(t_M-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_M)}(t_M-s)^\alpha)B_{r(t_M)}u(s)\mathrm{d}s\\
=\;&\underbrace{\Pi_{j=M}^{1}\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)(I+E_{r(t_{j-1})})}_{W_0}x_0\\
&+\sum_{i=1}^{M-1}\underbrace{\Pi_{j=M}^{i+1}\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)(I+E_{r(t_{j-1})})}_{W_i}\\
&\times\int_{t_{i-1}}^{t_i}(t_i-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}(t_i-s)^\alpha)B_{r(t_i)}u(s)\mathrm{d}s\\
&+\int_{t_{M-1}}^{t_M}(t_M-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_M)}(t_M-s)^\alpha)B_{r(t_M)}u(s)\mathrm{d}s\\
&+\sum_{i=2}^{M-1}\Pi_{j=M}^{i+1}\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)(I+E_{r(t_{j-1})})\mathcal{E}_\alpha(A_{r(t_i)}h_i^\alpha)F_{r(t_{i-1})}u_{i-1}\\
&+\mathcal{E}_\alpha(A_{r(t_M)}h_M^\alpha)F_{r(t_{M-1})}u_{M-1}\\
=\;&W_0x_0+\sum_{i=1}^{M}W_i\int_{t_{i-1}}^{t_i}(t_i-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}(t_i-s)^\alpha)B_{r(t_i)}u(s)\mathrm{d}s\\
&+\sum_{i=1}^{M-1}W_{i+1}\mathcal{E}_\alpha(A_{r(t_{i+1})}h_{i+1}^\alpha)F_{r(t_i)}u_i,
\end{aligned}
$$

where $W_M=I$ and $E_{r(t_0)}=0$.

3.2. Controllability criteria of FOSISs

In this subsection, the controllability of system (2.1) on $[t_0,t_M]$ will be investigated based on the obtained solution in the above subsection. The controllability definition is given as follows.

Definition 3.1. System (2.1) is (completely) controllable with regard to the sequence $\Sigma:=\{t_i\}_{i=1}^M$, if given any $x_0,x_f\in\mathbb{R}^n$, there exists a switching function $r(t)$ and a piecewise continuous input signal $u(t):[t_0,t_M]\to\mathbb{R}^m$ such that (2.1) satisfies $x(t_0)=x_0$ and $x(t_M)=x_f$.

Remark 3.2. It is worth pointing out that, in the classical linear continuous system theory [44], controllability to the origin (system is controlled from any state to the origin, i.e., $x(t_0)=x_0$ and $x(t_M)=0$ in Definition 3.1) is equivalent to reachability (system is controlled from the origin to any state, i.e., $x(t_0)=0$ and $x(t_M)=x_f$ in Definition 3.1). However, for impulsive systems, the above two notions are not equivalent [24,45], which can also be verified by the following example. In this paper, we investigate a more general controllability notion, i.e., complete controllability (system is controlled from any state to any state). It can be seen that, for systems with impulsive effects, controllability to the origin and reachability are only the special cases of complete controllability.

Example 3.1. Consider the controllability of a fractional-order impulsive system on a finite interval as follows:

$$
\begin{aligned}
{}^{C}_{t_{k-1}}D_t^\alpha x(t)&=Ax(t),\quad t\in[t_0,t_3]\backslash\{t_1,t_2\},\\
\Delta x(t_1)&=E_1x(t_1)+F_1u(t_1),\\
\Delta x(t_2)&=E_2x(t_2)+F_2u(t_2),\\
x(t_0)&=x_0,
\end{aligned}\tag{3.3}
$$

where $k = 1, 2, 3$, $t_i = i$ $(i = 0, 1, 2, 3)$, $n = 2$, $A = I$, $E_1 = 0$, $F_1 = \mathrm{diag}(2, 1)$, $E_2 = \mathrm{diag}(0, 1)$, $F_2 = 0$, $\alpha = 1/2$, $h_1 = h_2 = h_3 = 1$. Simple computations obtain

$$x(t) = \begin{cases} \mathcal{E}_\alpha(At^\alpha)x_0, \ t \in [0, 1], \\ \mathcal{E}_\alpha(A(t-1)^\alpha)\left((I+E_1)\mathcal{E}_\alpha(Ah_1^\alpha)x_0 + F_1 u(t_1)\right), \ t \in (1, 2], \\ \mathcal{E}_\alpha(A(t-2)^\alpha)\left(\Pi_{i=2}^{1}(I+E_i)\mathcal{E}_\alpha(Ah_i^\alpha)x_0 + (I+E_2)\mathcal{E}_\alpha(Ah_2^\alpha)F_1 u(t_1) + F_2 u(t_2)\right), \ t \in (2, 3]. \end{cases}$$

For any initial state x_0, one can always find a control:

$$u(t) = \begin{cases} -\mathrm{diag}\left(\frac{1}{2}\mathcal{E}_\alpha(1), \mathcal{E}_\alpha(1)\right)x_0, & t = 1 \\ 0, & t \in [0, 3]\backslash\{1\}. \end{cases}$$

By the explicit expression of $x(t)$, $x(3) = 0$ is finally derived. However, the second state variable cannot reach other points except the origin. Hence, system (3.3) is controllable to the origin with respect to $\Sigma := \{1, 2, 3\}$ while it is not reachable. This also implies that controllability to the origin is not equivalent to complete controllability.

Theorem 3.1. *System (2.1) is controllable with regard to $\Sigma := \{t_i\}_{i=1}^M$ iff there exists a switching function $r(t)$ satisfying* $\mathrm{rank}(W_c[t_0, t_1], W_c[t_1, t_2], \ldots, W_c[t_{M-1}, t_M], U_1, U_2, \ldots, U_{M-1}) = n$, *where*

$$\begin{aligned} W_c[t_{i-1}, t_i] = & \int_{t_{i-1}}^{t_i} (t_i - s)^{\alpha-1} W_i \mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}(t_i - s)^\alpha) B_{r(t_i)} \\ & \times B_{r(t_i)}^T \mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}^T (t_i - s)^\alpha) W_i^T \mathrm{d}s, \quad i = 1, 2, \ldots, M, \\ U_j = & W_{j+1}\mathcal{E}_\alpha(A_{r(t_{j+1})}h_{j+1}^\alpha)F_{r(t_j)}F_{r(t_j)}^T \mathcal{E}_\alpha(A_{r(t_{j+1})}^T h_{j+1}^\alpha)W_{j+1}^T, \quad j = 1, 2, \ldots, M-1. \end{aligned}$$

Proof. Denote $\hat{W} = \sum_{i=1}^M W_c[t_{i-1}, t_i] + \sum_{j=1}^{M-1} U_j$. It should be noticed that $W_c[t_{i-1}, t_i]$ and U_j in the form of the above equation are positive semidefinite. Hence, $\mathrm{rank}(\hat{W}) = \mathrm{rank}(W_c[t_0, t_1], W_c[t_1, t_2], \ldots, W_c[t_{M-1}, t_M], U_1, U_2, \ldots, U_{M-1})$ from Lemma 2.2. For the sufficiency, we only need to prove that there exists a piecewise continuous control function $u(t)$ such that

$$\begin{aligned} x_f - W_0 x_0 = & \sum_{i=1}^M W_i \int_{t_{i-1}}^{t_i} (t_i - s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}(t_i - s)^\alpha)B_{r(t_i)}u(s)ds \\ & + \sum_{i=1}^{M-1} W_{i+1}\mathcal{E}_\alpha(A_{r(t_{i+1})}h_{i+1}^\alpha)F_{r(t_i)}u_i. \end{aligned} \tag{3.4}$$

In fact, one can construct $u(t)$ as follows

$$u(t) = \begin{cases} B_{r(t_i)}^T \mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}^T (t_i - t)^\alpha) W_i^T \delta, \ t \in (t_{i-1}, t_i), \quad i = 1, 2, \ldots, M \\ F_{r(t_j)}^T \mathcal{E}_\alpha(A_{r(t_{j+1})}^T h_{j+1}^\alpha) W_{j+1}^T \delta, \quad t = t_j, \quad j = 1, 2, \ldots, M-1, \end{cases} \tag{3.5}$$

where $\delta \in \mathbb{R}^n$ needs to be designed. Applying the above control to Eq. (3.4), then one can obtain

$$\begin{aligned} x_f - W_0 x_0 = & \sum_{i=1}^M W_i \int_{t_{i-1}}^{t_i} (t_i - s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}(t_i - s)^\alpha)B_{r(t_i)} \\ & \times B_{r(t_i)}^T \mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}^T (t_i - t)^\alpha) W_i^T \delta \mathrm{d}s \\ & + \sum_{i=1}^{M-1} W_{i+1}\mathcal{E}_\alpha(A_{r(t_{i+1})}h_{i+1}^\alpha)F_{r(t_i)}F_{r(t_i)}^T \mathcal{E}_\alpha(A_{r(t_{i+1})}^T h_{i+1}^\alpha)W_{i+1}^T \delta \\ = & \sum_{i=1}^M W_c[t_{i-1}, t_i]\delta + \sum_{i=1}^{M-1} U_i \delta \\ = & \ \hat{W}\delta. \end{aligned} \tag{3.6}$$

One notes that there exists a unique δ satisfying the above equation when $\hat{W}$ has a full rank. Thus, the constructed control $u(t)$ is well designed. Therefore, we prove the sufficiency.

For the necessity, one can prove it by adopting the contradiction. Assume that $\mathrm{rank}(W_c[t_0, t_1], W_c[t_1, t_2], \cdots, W_c[t_{M-1}, t_M], U_1, U_2, \ldots, U_{M-1}) < n$. Thus, there is a nonzero $\bar{x} \in \mathbb{R}^n$ satisfying $\bar{x}^T W_c[t_0, t_1]\bar{x} = \cdots = \bar{x}^T W_c[t_{M-1}, t_M]\bar{x} =$

$\bar{x}^T U_1 \bar{x} = \cdots = \bar{x}^T U_{M-1} \bar{x} = 0$. Therefore, one has

$$\begin{aligned} &\bar{x}^T W_i \mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}(t_i - s)^{\alpha}) B_{r(t_i)} = 0, \quad s \in (t_{i-1}, t_i), \quad i = 1, 2, \ldots, M, \\ &\quad \bar{x}^T W_{j+1} \mathcal{E}_{\alpha}(A_{r(t_{j+1})} h_{j+1}^{\alpha}) F_{r(t_j)} = 0, \quad j = 1, 2, \ldots, M-1. \end{aligned} \tag{3.7}$$

Since the system is controllable with respect to $\Sigma := \{t_i\}_{i=1}^M$, it can be driven from a given initial point $x_0 \in \mathbb{R}^n$ to any point $x_f \in \mathbb{R}^n$. In particular, when $x_0 = 0$ and $x_f = \bar{x}(\bar{x} \neq 0)$, one can find a piecewise continuous control $\bar{u}(t)$ on $[t_0, t_M]$ satisfying

$$\begin{aligned} \bar{x} = &\sum_{i=1}^{M} W_i \int_{t_{i-1}}^{t_i} (t_i - s)^{\alpha-1} \mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}(t_i - s)^{\alpha}) B_{r(t_i)} \bar{u}(s) \mathrm{d}s \\ &+ \sum_{j=1}^{M-1} W_{j+1} \mathcal{E}_{\alpha}(A_{r(t_{j+1})} h_{j+1}^{\alpha}) F_{r(t_j)} \bar{u}_j. \end{aligned} \tag{3.8}$$

Multiplying $\bar{x}^T$ on (3.8), one obtains

$$\begin{aligned} \bar{x}^T \bar{x} = &\sum_{i=1}^{M} \bar{x}^T W_i \int_{t_{i-1}}^{t_i} (t_i - s)^{\alpha-1} \mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}(t_i - s)^{\alpha}) B_{r(t_i)} \bar{u}(s) \mathrm{d}s \\ &+ \sum_{j=1}^{M-1} \bar{x}^T W_{j+1} \mathcal{E}_{\alpha}(A_{r(t_{j+1})} h_{j+1}^{\alpha}) F_{r(t_j)} \bar{u}_j. \end{aligned}$$

From Eq. (3.7), one further derives $\bar{x}^T \bar{x} = 0$, which implies that $\bar{x} = 0$. This is a contradiction. This completes the necessity part. Thus, Theorem 3.1 is proved. ■

Remark 3.3. Different from the form of Gramian matrix in [46], the Gramian matrix $W_c[t_{i-1}, t_i]$ in continuous time is constructed including $(t_i - s)^{\alpha-1}$ in Theorem 3.1. This class of Gramian matrix has also been introduced in the existing literature for some different fractional systems, e.g., fractional systems with delay [11], fractional damped systems [12], and fractional systems with impulse [15]. In fact, if

$$W_c[t_{i-1}, t_i] = \int_{t_{i-1}}^{t_i} W_i \mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}(t_i - s)^{\alpha}) B_{r(t_i)} B_{r(t_i)}^T \mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}^T (t_i - s)^{\alpha}) W_i^T \mathrm{d}s,$$

that is $(t_i - s)^{\alpha-1}$ is not considered in the construction of the Gramian matrix, then Theorem 3.1 also holds.

Remark 3.4. In the study of controllability, to prove that a system is controllable, the existence of a desired controller needs to be verified. In fact, for FOSISs, a desired controller (i.e., Eq. (3.5)) that is dependent on a vector to be designed can be constructed according to the integral expression of the solution. Thus, the existence of a desired controller can be converted into the existence of a vector (i.e., δ in Eq. (3.5)). This implies that the existence of δ ensure the controllability. On other hand, for the controllability realization, it is in fact the designing problem of the controller. That is, one can design a desired controller that can satisfy any given initial and terminal states. Even if the controllability implies that the existence of a desired controller can be ensured, designing the desired controller cannot be ensured. For FOSISs, one can design the controller by obtaining the value of δ according to Eq. (3.6). Since FOSIS (2.1) is in fact a LTI system, the role of δ has similar effect for a general LTI system. That is, the existence of δ ensures the controllability and obtaining the value of δ can ensure the controllability realization.

Remark 3.5. It is noted that the derived condition in Theorem 3.1 is different from some existing results [24,25,28,39,40] for integer order impulsive and/or switching systems even if it can be reduced to the case of $\alpha = 1$. The differences can be summarized as follows. First, it can be found that there exist no extra assumptions about the impulsive matrices $(I+E_{r(t)})$ in Theorem 3.1, while the impulsive matrices are required to be nonsingular in [25,39,40]. Second, in integer order systems, the state transition matrix has some useful properties (e.g., $\Phi(t, t_0) = \Phi(t, \tau)\Phi(\tau, t_0)$ is satisfied for traditional state transition matrix $\Phi(t, t_0)$). These properties can be used to obtain the controllability conditions for integer order impulsive systems [39,40], which can be treated as a reduced case of our model. Different from the traditional state transition matrix, the Mittag-Leffler matrix functions are used to establish the Gramian matrices. However, the above mentioned properties, e.g., the transitive property ($\Phi(t, t_0) = \Phi(t, \tau)\Phi(\tau, t_0)$) is not satisfied here for the Mittag-Leffler matrix functions. Thus, a row matrix of several Gramian matrices is constructed to avoid using this property. Third, the Gramian matrices ($W_c[t_{i-1}, t_i]$ and U_j) in Theorem 3.1 consider both the continuous control and discrete control. However, in literature [25], the controllability results only consider the continuous control or discrete control. Thus, although our criteria are reduced to the integer order case, they still cover the existing results.

Corollary 3.1. *System (2.1) is controllable on the interval $[t_0, t_M]$ with regard to $\Sigma := \{t_i\}_{i=1}^M$ iff there exists a switching function $r(t)$ satisfying* $\mathrm{rank}(\Omega_1, \Omega_2, \ldots, \Omega_M, \Psi_1, \Psi_2, \ldots, \Psi_{M-1}) = n$, *where*

$$\begin{aligned}\Omega_i &= W_i R_{A_{r(t_i)},B_{r(t_i)}}, \quad i = 1, 2, \ldots, M,\\ \Psi_j &= W_{j+1}\mathcal{E}_\alpha(A_{r(t_{j+1})}h_{j+1}^\alpha)F_{r(t_j)}, \quad j = 1, 2, \ldots, M-1,\\ R_{A_{r(t_i)},B_{r(t_i)}} &= [B_{r(t_i)}, A_{r(t_i)}B_{r(t_i)}, \ldots, A_{r(t_i)}^{n-1}B_{r(t_i)}], \quad i = 1, 2, \ldots, M.\end{aligned}$$

Proof. We first prove the necessity. That is to show that if system (2.1) is controllable with respect to $\Sigma := \{t_i\}_{i=1}^M$, then the rank condition holds. If not so, i.e., $\mathrm{rank}(\Omega_1, \Omega_2, \ldots, \Omega_M, \Psi_1, \Psi_2, \ldots, \Psi_{M-1}) < n$, we can obtain a nonzero $x_\alpha \in \mathbb{R}^n$ satisfying $x_\alpha^T(\Omega_1, \Omega_2, \ldots, \Omega_M, \Psi_1, \Psi_2, \ldots, \Psi_{M-1}) = 0$, that is

$$\begin{aligned}x_\alpha^T W_i A_{r(t_i)}^j B_{r(t_i)} &= 0, \quad j = 0, \ldots, n-1, \quad i = 1, \ldots, M,\\ x_\alpha^T W_{j+1}\mathcal{E}_\alpha(A_{r(t_{j+1})}h_{j+1}^\alpha)F_{r(t_j)} &= 0, \quad j = 1, 2, \ldots, M-1.\end{aligned} \tag{3.9}$$

Based on Lemma 2.1, one knows $E_{\alpha,\alpha}(At^\alpha) = \sum_{i=0}^{n-1} f_i(t)A^i$, where $f_i(t)$ is a continuous function on $[0, \infty)$ for $i = 0, 1, 2, \ldots, n-1$. Hence, from (3.9), one has

$$\begin{aligned}x_\alpha^T W_c[t_{i-1}, t_i] &= \int_{t_{i-1}}^{t_i} (t_i - s)^{\alpha-1} x_\alpha^T W_i \mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}(t_i - s)^\alpha)B_{r(t_i)}\\ &\quad \times B_{r(t_i)}^T \mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}^T(t_i - s)^\alpha)W_i^T \mathrm{d}s\\ &= \int_{t_{i-1}}^{t_i} (t_i - s)^{\alpha-1} x_\alpha^T W_i \sum_{j=0}^{n-1} f_{i,j}(t_i - s)A_{r(t_i)}^j B_{r(t_i)}\\ &\quad \times B_{r(t_i)}^T \mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}^T(t_i - s)^\alpha)W_i^T \mathrm{d}s\\ &= \int_{t_{i-1}}^{t_i} (t_i - s)^{\alpha-1} \sum_{j=0}^{n-1} f_{i,j}(t_i - s)x_\alpha^T W_i A_{r(t_i)}^j B_{r(t_i)}\\ &\quad \times B_{r(t_i)}^T \mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}^T(t_i - s)^\alpha)W_i^T \mathrm{d}s\\ &= 0,\end{aligned}$$

where $i = 1, 2, \ldots, M$. Moreover, following (3.9), $x_\alpha^T U_j = x_\alpha^T W_{j+1}\mathcal{E}_\alpha(A_{r(t_{j+1})}h_{j+1}^\alpha)F_{r(t_j)}F_{r(t_j)}^T\mathcal{E}_\alpha(A_{r(t_{j+1})}^T h_{j+1}^\alpha)\ W_{j+1}^T = 0$ for $j = 1, 2, \ldots, M-1$. Thus, $\mathrm{rank}(W_c[t_0, t_1], W_c[t_1, t_2], \cdots, W_c[t_{M-1}, t_M], U_1, \ldots, U_{M-1}) < n$, which contradicts Theorem 3.1. This implies that the condition of $\mathrm{rank}(\Omega_1, \Omega_2, \ldots, \Omega_M, \Psi_1, \Psi_2, \ldots, \Psi_{M-1}) = n$ holds.

For the sufficient part, the aim is to prove that if $\mathrm{rank}(\Omega_1, \Omega_2, \ldots, \Omega_M, \Psi_1, \Psi_2, \ldots, \Psi_{M-1}) = n$, then $\mathrm{rank}(W_c[t_0, t_1], W_c[t_1, t_2], \ldots, W_c[t_{M-1}, t_M], U_1, \ldots, U_{M-1}) = n$. In fact, if not satisfied, there is a nonzero $\bar{x} \in \mathbb{R}^n$ satisfying $\bar{x}^T(W_c[t_0, t_1], W_c[t_1, t_2], \cdots, W_c[t_{M-1}, t_M], U_1 \cdots, U_{M-1}) = 0$.

On the one hand, $\bar{x}^T W_c[t_{i-1}, t_i]\bar{x} = 0$ implies that

$$\begin{aligned}&\bar{x}^T W_i \mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}(t_i - s)^\alpha)B_{r(t_i)}\\ &= \sum_{j=0}^{n-1} f_{i,j}(t_i - s)\bar{x}^T W_i A_{r(t_i)}^j B_{r(t_i)}\\ &= 0,\end{aligned}$$

where $s \in (t_{i-1}, t_i)$, $i = 1, 2, \ldots, M$. Denote $\bar{x}^T W_i A_{r(t_i)}^j B_{r(t_i)} = [\epsilon_{i,j,1}, \epsilon_{i,j,2}, \ldots, \epsilon_{i,j,d}]$, where $i = 1, \ldots, M, j = 0, 1, \ldots, n-1$. Thus, one obtains

$$\begin{cases}\sum_{j=0}^{n-1} f_{i,j}(t_i - s)\epsilon_{i,j,1} = 0\\ \vdots\\ \sum_{j=0}^{n-1} f_{i,j}(t_i - s)\epsilon_{i,j,d} = 0.\end{cases}$$

Note that $f_{i,j}(t)$ is linear independently from Lemma 2.1. Therefore, $\epsilon_{i,j,1} = \cdots = \epsilon_{i,j,d} = 0$. That is $\bar{x}^T W_i A_{r(t_i)}^j B_{r(t_i)} = 0$.

On the other hand, one has $\bar{x}^T W_{j+1}\mathcal{E}_\alpha(A_{r(t_{j+1})}h_{j+1}^\alpha)F_{r(t_j)} = 0$ from $\bar{x}^T U_j \bar{x} = 0$ for $j = 1, \ldots, M-1$. Then, one has $\bar{x}^T(\Omega_1, \Omega_2, \ldots, \Omega_M, \Psi_1, \Psi_2, \ldots, \Psi_{M-1}) = 0$, which implies that $\mathrm{rank}(\Omega_1, \Omega_2, \ldots, \Omega_M, \Psi_1, \Psi_2, \cdots, \Psi_{M-1}) < n$. This is a contradiction. This completes this proof. ■

Remark 3.6. The controllability condition in Corollary 3.1 is derived according to the Gramian matrices that are dependent on the Mittag-Leffler matrix function in Theorem 3.1. It is difficult to obtain the controllability condition if one directly uses some relevant matrix theory for the Gramian matrices. For this, a useful property (Lemma 2.1) about Mittag-Leffler matrix function is firstly established to simplify the controllability condition in Theorem 3.1. Based on this, we obtain the Kalman-type controllability condition.

Corollary 3.2. *Assume that $A_i = A, B_i = B, F_i = F$ and $AE_i = E_iA$ in system (2.1), where $i = 1, 2, \ldots, q$. Then for (2.1), a necessary controllability criterion with regard to $\Sigma := \{t_i\}_{i=1}^M$ is that there exists a switching function $r(t)$ satisfying*

$$\mathrm{rank}[G_1R_{A,B}, G_2R_{A,B}, \ldots, G_MR_{A,B}, G_2R_{A,F}, G_2R_{A,F}, \ldots, G_MR_{A,F}] = n,$$

where $G_i = \Pi_{j=M}^{i+1}(I+E_{r(t_{j-1})})(i = 1, 2, \ldots, M-1)$, $G_M = I$, $R_{A,B} = [B, AB, A^2B, \ldots, A^{n-1}B]$ and $R_{A,F} = [F, AF, A^2F, \ldots, A^{n-1}F]$. In particular, if $E_i = e_iI(e_i \neq -1, i = 1, 2, \ldots, q)$, the necessary condition is reduced to $\mathrm{rank}[R_{A,B}, R_{A,F}] = n$.

Proof. When $A_{r(t_i)} = A$, $AE_{r(t_i)} = E_{r(t_i)}A$, one has

$$W_i = \Pi_{j=M}^{i+1}\mathcal{E}_\alpha(Ah_j^\alpha)(I + E_{r(t_{j-1})}) = G_i\sum_{j=0}^{n-1}c_{i,j}A^j,$$

$$W_{i+1}\mathcal{E}_\alpha(Ah_{i+1}^\alpha) = G_{i+1}\sum_{j=0}^{n-1}d_{i,j}A^j,$$

where $i = 1, 2, \ldots, M-1$, $c_{i,j}$ and $d_{i,j}$ are fixed constants. Note that $\mathrm{Im}(W_{i+1}\mathcal{E}_\alpha(Ah_{i+1}^\alpha)F) = Im(G_{i+1}\sum_{j=0}^{n-1}d_{i,j}A^jF) \subseteq \mathrm{Im}\ (G_{i+1}R_{A,F})$. Thus, $\mathrm{Im}(\Psi_1, \Psi_2, \ldots, \Psi_{M-1}) \subseteq \mathrm{Im}(G_2R_{A,F}, \ldots, G_MR_{A,F})$. Similarly, one obtains $\mathrm{Im}(\Omega_1, \Omega_2, \ldots, \Omega_M) \subseteq \mathrm{Im}(G_1R_{A,B}, \ldots, G_MR_{A,B})$. Hence, $\mathrm{rank}(\Omega_1, \Omega_2, \ldots, \Omega_M, \Psi_1, \Psi_2, \ldots, \Psi_{M-1}) \leq \mathrm{rank}\,(G_1R_{A,B}, \ldots, G_MR_{A,B}, G_2R_{A,F}, \cdots, G_MR_{A,F}) \leq n$. From Corollary 3.1, if (2.1) is controllable regarding $\Sigma := \{t_i\}_{i=1}^M$, then $\mathrm{rank}(\Omega_1, \Omega_2, \ldots, \Omega_M, \Psi_1, \Psi_2, \ldots, \Psi_{M-1}) = n$, which implies that $\mathrm{rank}(G_1R_{A,B}, \ldots, G_MR_{A,B}, G_2R_{A,F}, \ldots, G_MR_{A,F}) = n$. This completes this proof. ■

Remark 3.7. In Corollary 3.2, the controllability result is mainly based on the controllability matrix in Corollary 3.1. When the system matrices satisfy some special constraints, by using Cayley–Hamilton theorem, the Mittag-Leffler matrix in Corollary 3.1 can be transformed into a form of matrix summation without including α. Thus, the obtained controllability condition is not related to the fractional order α. However, the result is proved to be the necessary but not sufficient controllability condition. Similarly, the observability result in the latter Corollary 3.4 is also necessary but not sufficient.

3.3. Observability criteria of FOSISs

Subsequently, another important property, i.e., observability of system (2.1) will be studied. Before obtaining the main results, the observability definition will be firstly given.

Definition 3.2. System (2.1) is (completely) observable with regard to the sequence $\Sigma := \{t_i\}_{i=1}^M$, if there exists a switching function $r(t)$ satisfying that the initial state $x_0 \in \mathbb{R}^n$ can be uniquely recovered by the known control input $u(t)$ and measured output $y(t)$ over the interval $[t_0, t_M]$.

Remark 3.8. A notion related to observability is called reconstructability. Different from observability definition, the reconstructability means that the terminal state $x(t_M) \in \mathbb{R}^n$ can be uniquely recovered by the known control input $u(t)$ and measured output $y(t)$ over the interval $[t_0, t_M]$ [44,45]. It is in fact a weak version of observability. Thus, the established observability results in the subsequent can be used to examine the reconstructability of system (2.1). However, the following example will show that a reconstructable system need not be an observable system.

Example 3.2. Consider a fractional-order impulsive system as follows:

$$\begin{aligned}
{}_{t_{k-1}}^{C}D_t^\alpha x(t) &= Ax(t), \quad t \in [t_0, t_3]\backslash\{t_1, t_2\},\\
\Delta x(t_1) &= E_1x(t_1),\\
\Delta x(t_2) &= E_2x(t_2),\\
x(t_0) &= x_0,\\
y(t) &= Cx(t),
\end{aligned}$$

where the system parameters are given as follows: $k = 1, 2, 3$, $t_i = i\ (i = 0, 1, 2, 3)$, $h_1 = h_2 = h_3 = 1$. Computations obtain that

$$x(t_3) = \mathcal{E}_\alpha(A)\Pi_{i=2}^1[(I + E_i)\mathcal{E}_\alpha(A)]x_0.$$

It can be seen that if this system is observable over [0,3], that is x_0 can be uniquely recovered by the output $y(t)$ on [0,3], then the terminal state $x(t_3)$ can also be obtained. This means that this system is also reconstructable. On the other hand, since $(I+E_1)$ and $(I+E_2)$ may be singular, $\mathcal{E}_\alpha(A)\Pi_{i=2}^{1}[(I+E_i)\mathcal{E}_\alpha(A)]$ may also be singular. Thus, the initial state x_0 cannot be uniquely obtained even if $x(t_3)$ can be uniquely recovered by the output $y(t)$ on [0,3] (i.e., this system is reconstructable). That is a reconstructable system may be not observable.

Theorem 3.2. *System (2.1) is observable with respect to $\Sigma := \{t_i\}_{i=1}^M$ iff there exists a switching function $r(t)$ satisfying rank$\left(\sum_{k=1}^M W_o[t_{k-1},t_k]\right)=n$ or rank $(W_o[t_0,t_1],W_o[t_1,t_2],\ldots,W_o[t_{M-1},t_M])=n$, where*

$$W_o[t_0,t_1]=\int_{t_0}^{t_1}\mathcal{E}_\alpha(A^T_{r(t_1)}(s-t_0)^\alpha)C^T_{r(t_1)}C_{r(t_1)}\mathcal{E}_\alpha(A_{r(t_1)}(s-t_0)^\alpha)\mathrm{d}s,$$
$$W_o[t_{k-1},t_k]=\int_{t_{k-1}}^{t_k}\left[\Pi_{j=k-1}^{1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)\right]^T\mathcal{E}_\alpha(A^T_{r(t_k)}(s-t_{k-1})^\alpha)C^T_{r(t_k)}$$
$$\times\ C_{r(t_k)}\mathcal{E}_\alpha(A_{r(t_k)}(s-t_{k-1})^\alpha)\Pi_{j=k-1}^{1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)\mathrm{d}s,\quad k=2,\ldots,M.$$

Proof. According to Lemma 3.1, it follows that

$$y(t)=C_{r(t_1)}\mathcal{E}_\alpha(A_{r(t_1)}(t-t_0)^\alpha)x_0$$
$$+\ C_{r(t_1)}\int_{t_0}^{t}(t-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_1)}(t-s)^\alpha)B_{r(t_1)}u(s)\mathrm{d}s+D_{r(t_1)}u(t),\ \ t\in[t_0,t_1],$$

and

$$y(t)=C_{r(t_k)}\mathcal{E}_\alpha(A_{r(t_k)}(t-t_{k-1})^\alpha)\left\{\Pi_{j=k-1}^{1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)x_0\right.$$
$$+\sum_{i=1}^{k-1}\Pi_{j=k-1}^{i+1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)(I+E_{r(t_i)})$$
$$\times\int_{t_{i-1}}^{t_i}(t_i-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}(t_i-s)^\alpha)B_{r(t_i)}u(s)\mathrm{d}s$$
$$\left.+\sum_{i=2}^{k-1}\Pi_{j=k-1}^{i}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)F_{r(t_{i-1})}u_{i-1}+F_{r(t_{k-1})}u_{k-1}\right\}$$
$$+\ C_{r(t_k)}\int_{t_{k-1}}^{t}(t-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_k)}(t-s)^\alpha)B_{r(t_k)}u(s)\mathrm{d}s+D_{r(t_k)}u(t),\ \ t\in(t_{k-1},t_k],\ \ k=2,\ldots,M.$$

We can rewrite the above equation as

$$\bar{y}(t)=\begin{cases}C_{r(t_1)}\mathcal{E}_\alpha(A_{r(t_1)}(t-t_0)^\alpha)x_0,\ \ t\in[t_0,t_1]\\ C_{r(t_k)}\mathcal{E}_\alpha(A_{r(t_k)}(t-t_{k-1})^\alpha)\Pi_{j=k-1}^{1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)x_0,\ \ t\in(t_{k-1},t_k],k=2,3,\ldots,M,\end{cases}$$

where

$$\bar{y}(t)=y(t)-C_{r(t_1)}\int_{t_0}^{t}(t-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_1)}(t-s)^\alpha)B_{r(t_1)}u(s)\mathrm{d}s-D_{r(t_1)}u(t),\ \ t\in[t_0,t_1]$$

and

$$\bar{y}(t)=y(t)-C_{r(t_k)}\mathcal{E}_\alpha(A_{r(t_k)}(t-t_{k-1})^\alpha)\left\{\sum_{i=1}^{k-1}\Pi_{j=k-1}^{i+1}(I+E_{r(t_j)})\right.$$
$$\times\ \mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)(I+E_{r(t_i)})\int_{t_{i-1}}^{t_i}(t_i-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_i)}(t_i-s)^\alpha)B_{r(t_i)}u(s)\mathrm{d}s$$
$$\left.+\sum_{i=2}^{k-1}\Pi_{j=k-1}^{i}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)F_{r(t_{i-1})}u_{i-1}+F_{r(t_{k-1})}u_{k-1}\right\}$$
$$-\ C_{r(t_k)}\int_{t_{k-1}}^{t}(t-s)^{\alpha-1}\mathcal{E}_{\alpha,\alpha}(A_{r(t_k)}(t-s)^\alpha)B_{r(t_k)}u(s)\mathrm{d}s-D_{r(t_k)}u(t),\ \ t\in(t_{k-1},t_k],k=2,\ldots,M.$$

It is easy to see, according to Definition 3.2, that the observability of (2.1) can be equivalently transformed into the estimation of x_0 by $\bar{y}(t)$. From the arbitrariness of $\bar{y}(t)$ and x_0, we need to obtain x_0 from $y(t)$ given by

$$y(t) = \begin{cases} C_{r(t_1)}\mathcal{E}_\alpha(A_{r(t_1)}(t-t_0)^\alpha)x_0, & t \in [t_0, t_1] \\ C_{r(t_k)}\mathcal{E}_\alpha(A_{r(t_k)}(t-t_{k-1})^\alpha)\Pi_{j=k-1}^{1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)x_0, & t \in (t_{k-1}, t_k], \quad k = 2, 3, \ldots, M \end{cases} \tag{3.10}$$

as $u(t) = 0$ and $u_k = 0$.

When $t \in [t_0, t_1]$, multiplying two sides of (3.10) by $\mathcal{E}_\alpha(A_{r(t_1)}^T(t-t_0)^\alpha)C_{r(t_1)}^T$. And when $t \in (t_{k-1}, t_k](k = 2, 3, \ldots, M)$, multiplying two sides of (3.10) by $\left[\Pi_{j=k-1}^{1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)\right]^T \mathcal{E}_\alpha(A_{r(t_k)}^T(t-t_{k-1})^\alpha)C_{r(t_k)}^T$. Thus, integrating with regard to t from t_0 to t_M obtains

$$\begin{aligned}
&\int_{t_0}^{t_1} \mathcal{E}_\alpha(A_{r(t_1)}^T(s-t_0)^\alpha)C_{r(t_1)}^T y(s)\mathrm{d}s \\
&\quad + \sum_{k=2}^{M}\int_{t_{k-1}}^{t_k}\left[\Pi_{j=k-1}^{1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)\right]^T \mathcal{E}_\alpha(A_{r(t_k)}^T(s-t_{k-1})^\alpha)C_{r(t_k)}^T y(s)\mathrm{d}s \\
&= \int_{t_0}^{t_1} \mathcal{E}_\alpha(A_{r(t_1)}^T(s-t_0)^\alpha)C_{r(t_1)}^T C_{r(t_1)}\mathcal{E}_\alpha(A_{r(t_1)}(s-t_0)^\alpha)\mathrm{d}s x_0 \\
&\quad + \sum_{k=2}^{M}\int_{t_{k-1}}^{t_k}\left[\Pi_{j=k-1}^{1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)\right]^T \mathcal{E}_\alpha(A_{r(t_k)}^T(s-t_{k-1})^\alpha)C_{r(t_k)}^T \\
&\quad \times C_{r(t_k)}\mathcal{E}_\alpha(A_{r(t_k)}(t-t_{k-1})^\alpha)\Pi_{j=k-1}^{1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)\mathrm{d}s x_0 \\
&= \left(\sum_{k=1}^{M} W_o[t_{k-1}, t_k]\right)x_0.
\end{aligned} \tag{3.11}$$

It can be seen that the left of (3.11) relies on $y(t)$, where $t \in [t_0, t_M]$. Hence, x_0 will be uniquely obtained from $y(t)$ if $\sum_{k=1}^{M} W_o[t_{k-1}, t_k]$ is invertible.

Conversely if $\operatorname{rank}\left(\sum_{k=1}^{M} W_o[t_{k-1}, t_k]\right) < n$, then there is a nonzero $x_\alpha \in \mathbb{R}^n$ satisfying

$$x_\alpha^T\left(\sum_{k=1}^{M} W_o[t_{k-1}, t_k]\right)x_\alpha = 0.$$

Choosing the initial state $x_0 = x_\alpha$, it yields from Eq. (3.10) that

$$\begin{aligned}
&\int_{t_0}^{t_M} y^T(s)y(s)\mathrm{d}s \\
&= \int_{t_0}^{t_1} x_\alpha^T\mathcal{E}_\alpha(A_{r(t_1)}^T(s-t_0)^\alpha)C_{r(t_1)}^T C_{r(t_1)}\mathcal{E}_\alpha(A_{r(t_1)}(t-t_0)^\alpha)x_\alpha \mathrm{d}s \\
&\quad + \sum_{k=2}^{M}\int_{t_{k-1}}^{t_k} x_\alpha^T\left[\Pi_{j=k-1}^{1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)\right]^T \mathcal{E}_\alpha(A_{r(t_k)}^T(s-t_{k-1})^\alpha)C_{r(t_k)}^T \\
&\quad \times C_{r(t_k)}\mathcal{E}_\alpha(A_{r(t_k)}(s-t_{k-1})^\alpha)\Pi_{j=k-1}^{1}(I+E_{r(t_j)})\mathcal{E}_\alpha(A_{r(t_j)}h_j^\alpha)x_\alpha \mathrm{d}s \\
&= x_\alpha^T\left(\sum_{k=1}^{M} W_o[t_{k-1}, t_k]\right)x_\alpha \\
&= 0,
\end{aligned}$$

which implies that $y(t) = 0$ for $t \in [t_0, t_M]$. Thus, from Definition 3.2, one knows that FOSIS (2.1) is not observable with regard to $\Sigma := \{t_i\}_{i=1}^{M}$. Besides, according to Lemma 2.2, $\operatorname{rank}\left(\sum_{k=1}^{M} W_o[t_{k-1}, t_k]\right) = n$ iff $\operatorname{rank}(W_o[t_0, t_1], W_o[t_1, t_2], \ldots, W_o[t_{M-1}, t_M]) = n$. Then the necessity is proved and this proof is finished. ■

Remark 3.9. It should be noted that the observability is a dual problem of controllability and hence the proposed observability conditions in Theorem 3.2 are established by using the method similar to controllability problem.

Remark 3.10. In Theorems 3.1 and 3.2, the integrals involve Mittag-Leffler matrix functions. Hence, the computing of the integrals relies on the solving of Mittag-Leffler matrix functions. In chapter 3.2 of literature [1], the authors provide several methods to calculate the value of the Mittag-Leffler matrix functions, e.g., inverse Laplace transform method, Jordan matrix decomposition method, and Cayley–Hamilton method.

Corollary 3.3. *System (2.1) is observable with regard to $\Sigma := \{t_i\}_{i=1}^M$ iff there exists a switching function $r(t)$ satisfying $\text{rank}(\psi_1, \psi_2, \ldots, \psi_M) = n$, where $\psi_1 = O_{A_{r(t_1)},C_{r(t_1)}}$, $\psi_k = \Pi_{j=1}^{k-1}\mathcal{E}_\alpha(A_{r(t_j)}^T h_j^\alpha)(I + E_{r(t_j)}^T)O_{A_{r(t_k)},C_{r(t_k)}}$, $k = 2, 3, \ldots, M$, and*

$$O_{A_{r(t_i)},C_{r(t_i)}} = \left[C_{r(t_i)}^T, A_{r(t_i)}^T C_{r(t_i)}^T, \ldots, (A_{r(t_i)}^T)^{n-1} C_{r(t_i)}^T\right],\ \ i = 1, 2, \ldots, M. \tag{3.12}$$

Proof. Please refer to the proof of Corollary 3.1. ■

Corollary 3.4. *Assume that $A_i = A, C_i = C$, and $AE_i = E_iA$ for $i = 1, 2, \ldots, q$ in system (2.1). Then for (2.1), a necessary observability criterion with regard to $\Sigma := \{t_i\}_{i=1}^M$ is that there exists a switching function $r(t)$ satisfying $\text{rank}(O_{A,C}, H_1O_{A,C}, H_2O_{A,C}, \ldots, H_{M-1}O_{A,C}) = n$, where $H_i = \Pi_{j=1}^{i}(I + E_{r(t_j)}^T)(i = 1, 2, \ldots, M-1)$, $O_{A,C} = [C^T, A^TC^T, (A^T)^2C^T, \ldots, (A^T)^{n-1}C^T]$. In particular, if $E_i = e_iI (e_i \neq -1, i = 1, 2, \ldots, q)$, the necessary condition reduces to $\text{rank}(O_{A,C}) = n$.*

Proof. Please refer to the proof of Corollary 3.2. ■

Remark 3.11. For a general LTI system, including integer order case (i.e., $\dot{x} = Ax + Bu$ and $y = Cx$) and fractional order case (i.e., ${}_0^CD_t^\alpha = Ax + Bu$ and $y = Cx$), the controllability and observability conditions have been verified to only be dependent on the system matrices. That is, the controllability condition is $\text{rank}(R_{A,B}) = n$ and the observability condition is $\text{rank}(O_{A,C}) = n$ for integer or fractional case. It can be seen that α plays no role in controllability and observability [1]. Later, for some other systems, e.g., fractional damped systems with delay [12], fractional impulsive systems [15], and fractional impulsive system with delay [13], the controllability and observability conditions that are not related to the fractional order α are also established. This is in fact a meaningful finding since it provide some more easily verified controllability and observability criteria compared to the condition in the form of Gramian matrix. In terms of our model (2.1), Theorems 3.1 and 3.2 establish some Gramian matrix conditions to test the controllability and observability and then these conditions are equivalently transformed into more concise controllability and observability matrices related to Mittag-Leffler matrix in Corollaries 3.1 and 3.3. Similar to the conclusion in the above mentioned fractional systems, an intuitive conjecture is that the rank of controllability and observability matrices in Theorems 3.1 and 3.2 and Corollaries 3.1 and 3.3 will not change with fractional order α. Hence, the aim in Corollaries 3.2 and 3.4 is to try to transform the conditions related to Mittag-Leffler matrix into some matrix conditions that are not relevant to fractional order α with the aid of Cayley–Hamilton theorem. However, it does not mean that the fractional order α has no effect on the controllability and observability for any system. In the existing literature [47], we note that for a discrete-time linear system with fractional different order, the effect of the fractional order on the observability and controllability has been presented. For more other systems, the importance of fractional order in the controllability and observability deserves to explore.

4. An illustrating example

In the following, the obtained controllability and observability criteria are tested by a numerical example.

Example 4.1. Consider a FOSIS with three different subsystems as follows:

$$\begin{aligned}
{}_{t_{k-1}}^{C}D_t^\alpha x(t) &= A_{r(t)}x(t) + B_{r(t)}u(t), \quad t \in [0, 3]\backslash\{t_k\}_{k=1}^2, \\
\Delta x(t) &= E_{r(t)}x(t) + F_{r(t)}u_k, \quad t = t_k, \\
x(t_0) &= x_0, \\
y(t) &= C_{r(t)}x(t),
\end{aligned}$$

where $\alpha = 0.5$, $t_i = i(i = 0, 1, 2, 3)$, and the related coefficient matrices are given by

$$A_1 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, B_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, A_2 = \begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix}, B_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, A_3 = \begin{bmatrix} 0 & 3 \\ 0 & 0 \end{bmatrix}, B_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \tag{4.13}$$

$$E_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad F_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad F_2 = \begin{bmatrix} 0 \\ 2 \end{bmatrix}, \quad F_3 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \tag{4.14}$$

$$C_1 = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad C_2 = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad C_3 = \begin{bmatrix} 0 & 1 \end{bmatrix}. \tag{4.15}$$

One notes that every subsystem in (4.13) is not controllable according to the classical Kalman criterion [1]. Next, we use Theorem 3.1 to illustrate the controllability of this example. For the given impulse-switching time sequence $\Sigma := \{t_i\}_{i=1}^3$, a switching function is chosen as $r(t) = i, t \in (t_{i-1}, t_i]$. By computations, one obtains

$$W_c[0, 1] = \begin{bmatrix} 1710.6 & 446.8 \\ 446.8 & 116.7 \end{bmatrix}, W_c[1, 2] = \begin{bmatrix} 65.6561 & 19.3954 \\ 19.3954 & 5.7296 \end{bmatrix}, W_c[2, 3] = \begin{bmatrix} 0.6366 & 0 \\ 0 & 0 \end{bmatrix},$$

$$U_1 = \begin{bmatrix} 671.7589 & 175.4741 \\ 175.4741 & 45.8366 \end{bmatrix}, U_2 = \begin{bmatrix} 45.8366 & 13.5406 \\ 13.5406 & 4.0000 \end{bmatrix}.$$

It is observed that rank($W_c[0, 1], W_c[1, 2], W_c[2, 3], U_1, U_2$)=2. Thus, the above system is controllable with respect to $\Sigma := \{t_i\}_{i=1}^3$ according to Theorem 3.1.

When the switching function is chosen as $r(t) = i$ for $t \in (t_{i-1}, t_i]$, according to the precise expressions of Ω_i and Ψ_j in Corollary 3.1, one can obtain the following

$$\Omega_1 = \begin{bmatrix} 51.8366 & 0 \\ 13.5406 & 0 \end{bmatrix}, \Omega_2 = \begin{bmatrix} 10.1554 & 0 \\ 3.0000 & 0 \end{bmatrix}, \Omega_3 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \Psi_1 = \begin{bmatrix} 25.9183 \\ 6.7703 \end{bmatrix}, \Psi_2 = \begin{bmatrix} 25.9183 \\ 6.7703 \end{bmatrix}.$$

It can be found that rank$(\Omega_1, \Omega_2, \Omega_3, \Psi_1, \Psi_2) = 2$, which implies that the above system is controllable according to Corollary 3.1.

In fact, for some constants $d_1, d_2 \in \mathbb{R}$, assume that $u(t) = d_1, t \in [0, 1), u(t) = 0, t \in [1, 2], u(t) = d_2, t \in (2, 3]$. Then, according to Lemma 3.1, $x(3) = \begin{bmatrix} 97.6732 & 20.3108 \\ 27.0811 & 6.0000 \end{bmatrix} x_0 + \begin{bmatrix} 58.4914 \\ 15.2789 \end{bmatrix} d_1 + \begin{bmatrix} 1.1284 \\ 0 \end{bmatrix} d_2$. Thus for any $x_0 \in \mathbb{R}^n$, $x(3)$ will be driven anywhere by choosing appropriate d_1 and d_2. Thus, the controllability of this system is achieved.

On the other hand, by the definition of $W_o[t_{i-1}, t_i](i = 1, 2, 3)$, one can obtain

$$W_o[0, 1] = \begin{bmatrix} 1.0000 & 0.7523 \\ 0.7523 & 0.6366 \end{bmatrix}, W_o[1, 2] = \begin{bmatrix} 26.2220 & 39.6063 \\ 39.6063 & 59.9951 \end{bmatrix}, W_o[2, 3] = \begin{bmatrix} 183.3465 & 288.1277 \\ 288.1277 & 452.7905 \end{bmatrix}.$$

Thus, rank($W_o[0, 1], W_o[1, 2], W_o[2, 3]$)=2, and hence, by Theorem 3.2, one concludes that this system is observable with respect to $\Sigma := \{t_i\}_{i=1}^3$. In fact, if $u(t) = 0$, according to Eq. (3.11), we can obtain the initial state x_0 from $y(t)$ on $[t_0, t_3]$. This implies that the above system is observable.

When the switching function is chosen as $r(t) = i$ for $t \in (t_{i-1}, t_i]$, according to the precise expressions of ψ_1, ψ_2, ψ_3 in Corollary 3.3, it follows that

$$\psi_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \psi_2 = \begin{bmatrix} 2.0000 & 4.0000 \\ 4.2568 & 4.5135 \end{bmatrix}, \psi_3 = \begin{bmatrix} 13.5406 & 0 \\ 21.2789 & 0 \end{bmatrix}.$$

It can be easily seen that rank$(\psi_1, \psi_2, \psi_3) = 2$. This also means that the above system is observable according to Corollary 3.3.

5. Conclusion

The problems of controllability and observability have been addressed for a kind of FOSISs. First, the general solution of FOSISs has been derived on all impulsive intervals according to the fractional-order Laplace theory and mathematical induction. Next, several necessary and sufficient controllability and observability criteria for such hybrid systems have been respectively established in the form of Gramian matrices based on the general solution expression and matrix theory. Furthermore, to reduce the complexity of computation, some easily tested rank conditions similar to the classical Kalman rank criterion for the controllability and observability have also been obtained. Finally, an example has been given to illustrate the theoretical results.

CRediT authorship contribution statement

Jiayuan Yan: Conceptualization, Methodology, Writing – original draft. **Bin Hu:** Conceptualization, Methodology, Writing – original draft. **Zhi-Hong Guan:** Conceptualization, Supervision, Resources, Investigation. **Tao Li:** Validation, Writing – review & editing. **Ding-Xue Zhang:** Validation, Writing – review & editing.

Declaration of competing interest

No potential conflict of interest was reported by the authors.

Data availability

No data was used for the research described in the article.

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